

ExoHabitAI - Milestone 4 Document

This document presents the details of the ExoHabitAI project, including system overview, working methodology, and application link.

1. Project Overview

ExoHabitAI is a machine learning-based web application designed to predict whether an exoplanet can support life. It uses astrophysical parameters like radius, mass, temperature, density, and star temperature.

2. Working Methodology

The system follows ML pipeline: preprocessing, feature engineering, label creation using scientific rules, model training with Random Forest, and deployment using Streamlit.

3. Features

- Habitability prediction using ML
- Interactive UI
- Confidence score
- Scientific explanation

4. Application Link

https://replit.com/stripe-checkout-by-price/core_1mo_20usd_monthly_feb_26?coupon=AGENT42B21F40B3C03

5. Technologies Used

Python, Pandas, NumPy, Scikit-learn, Streamlit, SMOTE, Random Forest